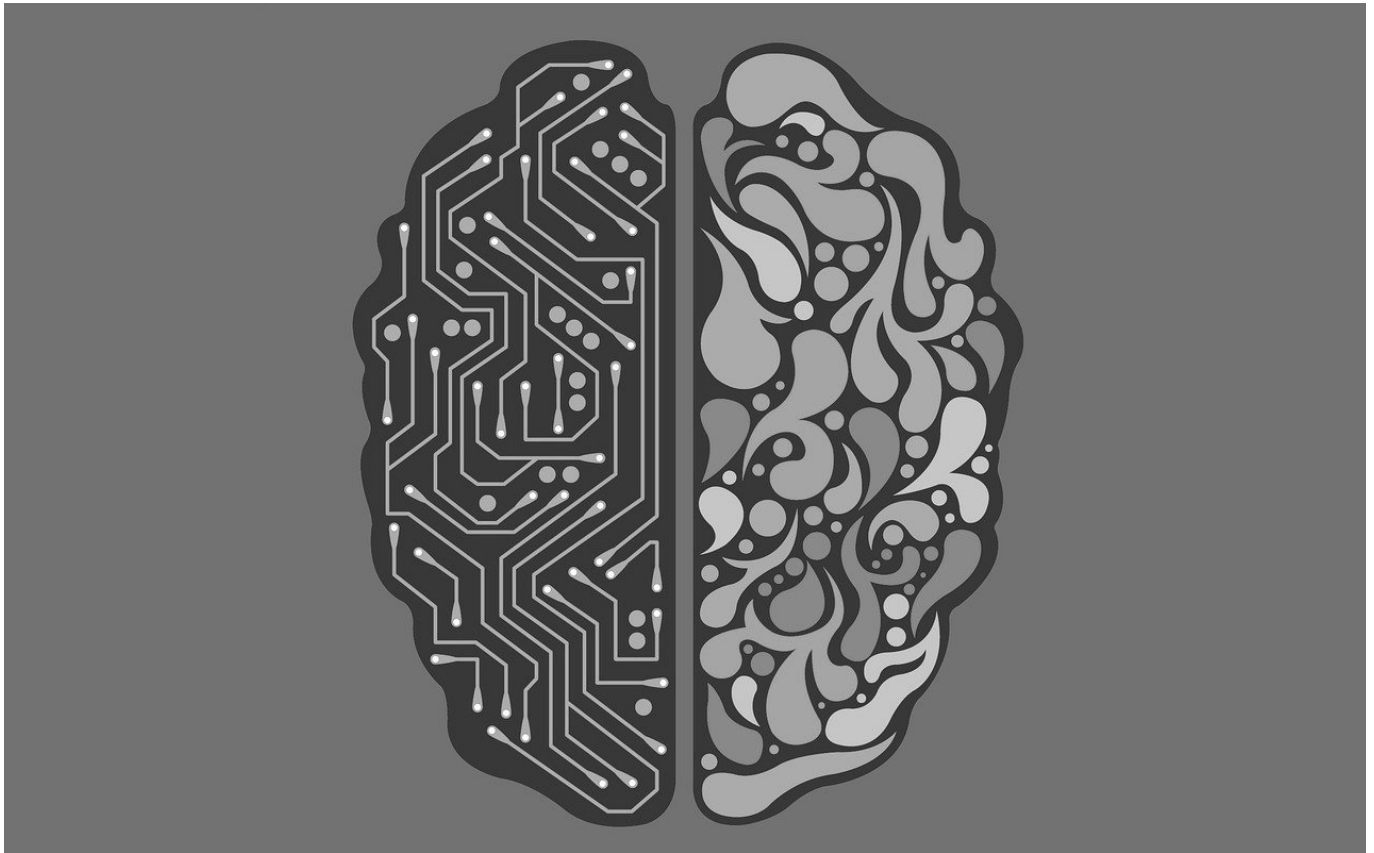


Sensors

Artificial Intelligence



Artificial Intelligence (AI) is a technology that allows machines to **learn, understand, and process information** like humans. One of the key components of AI systems is sensors.

Sensors are devices that can **detect and measure** physical and environmental conditions such as temperature, light, sound, and movement. In this lesson, we will learn about the different types of sensors used in AI applications and the different ways in which they can be used.

Types of sensors

- **Optical sensors:** These sensors detect light and are used in applications such as facial recognition, object detection, and image processing. Examples of optical sensors include cameras and lidar sensors.
- **Temperature sensors:** These sensors measure temperature and are used in applications such as climate control and food safety. Examples of temperature sensors include thermocouples and thermistors.
- **Pressure sensors:** These sensors measure pressure and are used in applications such as industrial automation, weather forecasting, and healthcare. Examples of pressure sensors include piezoelectric sensors and strain gauge sensors.

- **Accelerometer sensors:** These sensors measure acceleration and are used in applications such as motion detection, navigation, and gaming. Examples of accelerometer sensors include MEMS accelerometers and piezoelectric accelerometers.
- **Gyroscopic sensors:** These sensors measure angular velocity and are used in applications such as navigation, gaming, and robotics. Examples of gyroscopic sensors include MEMS gyroscopes and fiber optic gyroscopes.
- **Magnetic sensors:** These sensors measure magnetic fields and are used in applications such as navigation, industrial automation, and healthcare. Examples of magnetic sensors include Hall effect sensors and magnetoresistive sensors.
- **Ultrasonic sensors:** These sensors measure distance and are used in applications such as object detection, navigation, and industrial automation. Examples of ultrasonic sensors include sonar sensors and lidar sensors.
- **Infrared sensors:** These sensors detect infrared radiation and are used in applications such as temperature measurement, night vision, and gesture recognition. Examples of infrared sensors include thermopile sensors and pyroelectric sensors.
- **Proximity sensors:** These sensors detect the presence of objects and are used in applications such as gesture recognition, object detection, and access control. Examples of proximity sensors include infrared proximity sensors and ultrasonic proximity sensors.
- **Light sensors:** These sensors detect light and are used in applications such as light control, gesture recognition, and object detection. Examples of light sensors include photodiodes and phototransistors.
- **Humidity sensors:** These sensors measure humidity and are used in applications such as weather forecasting, agriculture, and healthcare. Examples of humidity sensors include capacitive humidity sensors and resistive humidity sensors.
- **Gas sensors:** These sensors detect the presence of gases and are used in applications such as environmental monitoring, industrial automation, and healthcare. Examples of gas sensors include electrochemical gas sensors and metal oxide gas sensors.

Images

1. [Artificial Intelligence](#)